

Sharad Patel

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sharadrpatel.com

Education

University of Florida, BS in Biomedical Engineering

Aug 2023 – May 2027

- GPA: 3.94/4.0

Research Experience

Undergraduate Research Assistant, Lung Modeling Lab (Dr. Brunson) –
Gainesville, FL

May 2026 – Present

- Developing a mechanistic model of primary graft dysfunction (PGD) in lung transplantation to characterize underlying pathophysiological drivers and inform risk stratification.
- Conducting a systematic review of the PGD literature to synthesize existing mechanistic and clinical findings and identify gaps to guide model development.

Undergraduate Research Assistant, Holt Lab – Gainesville, FL

Dec 2025 – Present

- Built a computer vision and ML pipeline using SAM to automatically segment leaf images, isolate regions of interest, and quantify infected tissue area.
- Computed disease severity metrics by estimating percent infection across samples to support large-scale ecological analysis.
- Constructed simulation models of environmental change incorporating phenotypic plasticity via sigmoidal response functions; implemented and validated ODE-based ecological models of population and disease dynamics.
- Extended the modeling framework to incorporate cell/organism migration dynamics, coupling movement behavior with plasticity responses to capture spatial spread patterns under environmental change.

Undergraduate Research Assistant, Vedam-Mai Lab – Gainesville, FL

Sept 2025 – Present

- Lead author on *Systematic Review: Immunotherapy in Parkinson's Disease* – designing the search strategy, screening and extracting studies, and synthesizing evidence across active, passive, and cell-based immunotherapy approaches.
- Investigating immune mechanisms in Parkinson's disease, focusing on T-cell responses and neuroinflammation along the gut-brain axis.
- Analyzing biological datasets to evaluate relationships between immune markers, microbiome composition, and disease progression; quantifying and visualizing results to identify inflammatory signaling trends.
- Contributing to experimental design discussions and preparation of research documentation, figures, and lab presentations.

Undergraduate Research Assistant, Dr. Nichols Lab – Gainesville, FL

Aug 2024 – Present

- Collected fine-wire EMG and motion capture data using Vicon software; processed signals in Python to evaluate movement patterns and muscle activation.
- Built subject-specific OpenSim thumb models personalized with ultrasound-derived muscle measurements; compared predicted muscle forces between healthy older adults and adults with carpometacarpal osteoarthritis (CMCOA).
- Applied OpenSim inverse kinematics and static optimization to analyze joint angles, moment arms, and muscle forces across pinch and grasp tasks.
- Conducted sensitivity and uncertainty analyses on ultrasound-derived Hill-type parameters (optimal fiber length, tendon slack length, physiological cross-sectional area) to identify dominant drivers of personalization and propagate measurement error into force predictions.
- Developed a reduced Hill-type musculoskeletal model to test whether simplified formulations reproduce the personalization effects of the full OpenSim simulation.

- Performed exploratory data analysis on multi-muscle EMG recordings, including paired *t*-tests and two-way ANOVA to assess activation differences across CMCOA stages.

Clinical Experience

Certified Nursing Assistant, HCA Florida North Florida Hospital – Gainesville, FL May 2026 – Present

- Provide direct patient care on a medical–surgical unit: vital signs, activities of daily living, ambulation and repositioning, intake/output, and blood glucose monitoring.
- Document care in the electronic health record and report changes in patient status to nursing staff; escalate acute changes promptly.
- Support admissions, transfers, and discharges on a high-turnover unit; assist with post-operative care, fall prevention, and infection control protocols.

Engineering & Leadership

Senior Design Team Member, CONMED Distal Biceps Tendon Repair System Aug 2026 – Present

- Designing a proof-of-concept distal biceps tendon repair system (implant, instruments, and sterile packaging) on an 8-person team for industry sponsor CONMED, for evaluation by a sports medicine surgeon in a cadaveric simulated-use setting.
- Co-developed the needs statement and design traceability matrix: 12 user needs with marginal/ideal target specifications grounded in biomechanical literature and FDA, ISO, and ASTM standards (e.g., ≥ 310 – 440 N load to failure after 3,600 physiologic cycles, ≤ 3 mm tendon–bone gap, ≥ 5 mm nerve clearance).
- Benchmarked cortical-button, suture-anchor, and bone-tunnel constructs and reviewed IP and regulatory requirements, identifying a Class II 510(k) pathway (21 CFR 888.3040) and implant cost targets.

Surgical Research Team Captain, Dream Team Engineering – Gainesville, FL Jan 2024 – Present

- Lead surgical simulation research: authored IRB applications, study protocols, and statistical analysis plans; designed and conducted skills assessments for surgical residents.
- Collaborated with transplant surgeons to develop realistic simulation models for kidney and liver transplant procedures and infant IV administration.
- Published findings from surveys and experimental data; contributed to team publications and conference presentations.

Circuits & Software Team Member, GRiP Gaming Team – Gainesville, FL Aug 2023 – May 2024

- Co-developed electrical and software systems for a 3D-printed prosthetic hand for a child with a below-elbow amputation.
- Programmed hand closure response to bicep myosensor signals using Arduino and C++.

Teaching Assistant – Cellular Systems & Physiology, University of Florida Jan 2026 – Present

- Lead weekly laboratory sessions guiding students through human physiology experiments; assist in applying engineering reasoning to interpret physiological data and build conceptual models.
- Rewrote and redesigned the course laboratory curriculum across the summer and fall semesters, restructuring activities and handouts around clearer learning objectives and stronger conceptual scaffolding.
- Revised lab procedures and assessment questions based on recurring student errors, improving clarity of instructions and alignment between lab work and lecture content.

Teaching Assistant – Organic Chemistry, University of Florida Jan 2026 – Aug 2026

- Conduct weekly office hours covering reaction mechanisms, synthesis strategy, and spectroscopy; grade assessments and provide targeted feedback to address recurring conceptual errors.

Learning Assistant – Biomedical Transport Phenomena, University of Florida Aug 2026 – Present

- Hold weekly office hours and lead exam review sessions, walking students through momentum, heat, and mass transport problem-solving and clarifying recurring conceptual difficulties.

Co-Director of Internal Events, Dream Team Engineering – Gainesville, FL May 2024 – May 2025

- Organized workshops, social events, and general body meetings; hosted large-scale events including a 24-Hour Design Challenge and a Shark Tank-style competition.

- Regional Core Team**, BAPS Swaminarayan Sanstha – Tampa, FL May 2021 – Present
- Design comprehensive programs for 750–1,000 children (ages 6–13); collaborate with a team of six to develop and deliver engaging weekly curriculum for 50–100 attendees.
- Project Lead**, BAPS Charities – Tampa, FL Aug 2022 – Present
- Led multi-cause community events including annual walkathons and health fairs; scoped goals and budgets, secured permits, coordinated 30–100+ volunteers, and managed sponsor outreach.
 - Planned and executed blood drives with local hospitals; handled logistics, screening workflows, and safety compliance to reduce wait times and increase throughput.
- Video Editor**, Festival of Inspirations – Robbinsville, NJ June 2023 – Aug 2023
- Scripted and produced video content for a three-month cultural festival dedicated to Hindu values and traditions.
- Math Tutor**, Mathnasium – Tampa, FL Aug 2022 – June 2023
- Provided tailored mathematics instruction to a weekly cohort of 50 K–12 students; designed personalized learning plans aligned to individual academic needs.

Publications & Presentations

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- Evaluation of a Mixed-Reality Infant Caudal Epidural Block Simulator for Procedural Training** Oct 2026
Sharad Patel, Pranav Kulkarni, Sanandan Ojha
Poster Presentation, 2026 BMES Annual Meeting – Orlando, FL
- Evaluation of a 3D Kidney Transplant Simulator for Surgical Education: A Study Protocol** Oct 2026
 Sanandan Ojha, Thien-Nhi Nguyen, Shreya Sreekanth, *Sharad Patel*
Poster Presentation, 2026 BMES Annual Meeting – Orlando, FL
- Improving Pediatric MRIs: A Prospective Study using a Mock Scanner as a Replacement for Sedation** Aug 2024
 Shreya Mathur, Utsav Sharma, Ansh Parikh, Daisy Ann Titus, *Sharad Patel*
International Undergraduate Journal of Medicine, Disease, and Society
- Effectiveness of 3D Box Model Trainers in Kidney Transplantation Training: A Controlled Study** Mar 2025
 John Cabrera, *Sharad Patel*, Makenna Myrick, Claudia Klejc, Ania Kelegama, Anne Titus
Presented at: 2024 Surgeons and Engineers: A Dialogue on Surgical Simulation
- Enhancing Anesthesia Monitoring with Computer Vision-Powered Train-of-Four Testing** Apr 2025
 Ansh Parikh, *Sharad Patel*, Vineet Khanolkar, Aditya Jithesh, Nishant Nagururu, Utsav Sharma
Poster Presentation, Engineering Research Symposium, PHHE
- Modeling AI Model Collapse using Markov Chains** Apr 2026
Sharad Patel, Pranav Kulkarni, Sanandan Ojha
Presented at: Undergraduate Mathematics Research Symposium, University of Florida

Projects

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- Mechanobiological Model of In-Stent Restenosis in Peripheral Stents** Aug 2026 – Present
- Adapting a published reaction–diffusion model of restenosis (Escuer et al., 2019), which couples growth factors, MMP-2, ECM, and contractile/synthetic smooth muscle cells, from balloon-expandable coronary stents to self-expanding nitinol stents in the femoral artery, on a 5-person team with Medtronic’s Modeling & Simulation Innovation Group.
 - Led problem definition and physiological rationale, connecting the stent’s chronic outward force to smooth muscle cell migration and proliferation, ECM deposition, and neointimal hyperplasia.
 - Added a chronic outward force module linking stent oversizing to arterial hoop stress. Benchmarked the re-implemented base model against published outputs (24.8% vs. ~25% restenosis at 300 days). In a blind test

against porcine data, the phenotype-switch pathway predicted 73% area stenosis (observed $64 \pm 17\%$).

- Tools Used: PDE modeling, numerical simulation, parameter calibration, sensitivity analysis

Height-Adjustable Axillary Crutch – CAD Design & Technical Report

Oct 2025

- Designed an 11-part telescoping pediatric crutch in Onshape, sized with the 77%-of-height clinical rule to fit users 1.33–1.56 m tall with ~ 180 mm of pin-locked height adjustment.
- Built a full assembly with fastened and slider mates and travel limits to simulate the telescoping leg and removable locking pins.
- Produced dimensioned multiview engineering drawings for every part, specifying ± 0.01 mm tolerances at sliding and pin interfaces and choosing materials for each part (aluminum alloy frame, ABS connector, TPU/rubber grips, stainless steel pins).
- Wrote a technical report covering design justification, part-by-part drawings, tolerancing rationale, and design limitations, with next steps including FEA load simulation and spring-loaded locking.
- Tools Used: Onshape, Fusion 360, tolerancing and technical drawing, technical writing

Personal Portfolio Website

sharadrpatel.com

- Designed and built a personal portfolio website to showcase research, projects, and publications.
- Implemented responsive UI/UX with modern web standards; deployed and maintained as a live production site.
- Tools Used: HTML, CSS, JavaScript, React

Connections Word Game – Full-Stack Web Application

github.com/sharadrpatel/
BKST_NYC

- Built a fully functional clone of the NYT Connections word puzzle game with a complete front-end and back-end implementation.
- Replicated core gameplay mechanics including animated tile interactions, color-coded category reveals, and guess tracking.
- Implemented a persistent global leaderboard tracking scores across sessions, enabling competitive play between users.
- Tools Used: JavaScript, HTML, CSS

COMAP MCM – Smartphone Battery Dynamics Model

- Developed a physics-grounded continuous-time ODE model predicting smartphone battery state-of-charge (SOC) under realistic user behavior, incorporating processor load, display brightness, network activity, and thermodynamic effects.
- Modeled user activity via stochastic indicator functions and Fourier-series CPU/network representations; performed global Monte Carlo sensitivity analysis (10^4 simulations) identifying processor load as the dominant depletion driver.
- Tools Used: MATLAB, ODE45, stochastic modeling, uncertainty quantification

SCUDEM International Modeling Competition – *Outstanding* Award

- Modeled AI model collapse dynamics using two complementary frameworks: a Markov-chain ecosystem model for early-stage information loss and a layered discrete distribution model for late-stage collapse.
- Simulated recursive human-AI training via Monte Carlo experiments; analyzed distribution drift using KL divergence and tail coverage metrics; demonstrated that model diversity stabilizes performance.
- Received *Outstanding* designation (highest award level) among international competing teams.
- Tools Used: MATLAB, probability theory, stochastic processes, information theory

Spotted Seatrout Population ODE Modeling & Management Simulator

github.com/sharadrpatel/
Fish_Stock_ODE_Modeling

- Built a stock-assessment ODE model for Florida spotted seatrout combining logistic growth with fishing-mortality removals; integrated FWC landing, CPUE, and SSB data (2000–2023) into a unified time series.
- Estimated population parameters via constrained Nelder–Mead optimization; validated trajectories against independent SSB data with $< 10\%$ relative error; simulated 10-year constant- F policies to identify F_{MSY} and B_{MSY} .

- Tools Used: Python (NumPy, pandas, SciPy, Matplotlib), Jupyter, Git

Neural Network from Scratch

- Implemented a feedforward neural network from first principles in NumPy (no deep learning frameworks), including forward propagation, backpropagation, and mini-batch gradient descent.
- Trained on MNIST handwritten digit dataset; visualized training/validation loss and accuracy curves.
- Tools Used: Python (NumPy, Matplotlib)

Leaf Segmentation Pipeline (SAM) for Disease Severity Quantification

- Built an end-to-end computer vision pipeline using SAM to segment leaves, generate per-leaf masks, and compute infection severity metrics including percent infected tissue and structural damage via contour and convex-hull analysis.
- Implemented reference-tag calibration to convert pixel measurements to real-world area units; optimized execution on Apple Silicon using PyTorch MPS acceleration.
- Tools Used: Python (OpenCV, NumPy), PyTorch, Ultralytics

Statistical Analysis of Student Organization Applications

- Analyzed multi-year application data to identify trends, acceptance predictors, and sources of reviewer bias; quantified reviewer-order effects on final outcomes.
- Built scoring dashboards and recommended process changes (blinded first-pass review, balanced reviewer assignment, calibration sessions) to reduce bias and improve fairness.
- Tools Used: Python (NumPy, pandas, Matplotlib/Seaborn)

Skills & Coursework

Programming: Python, MATLAB, C++ , R, JavaScript, LaTeX

Certifications: CNA, BLS

Tools & Techniques: SolidWorks, Onshape, Fusion 360, Arduino, OpenSim, Vicon, Git, 3D Printing, BSL-2 Lab Techniques, Scientific Writing

Relevant Coursework: Differential Equations, Linear Algebra, Modeling in Mathematical Biology, Biomedical Instrumentation, Bio Signals, Circuits, Biochemistry, Organic Chemistry, Genetics, Biomaterials, Cellular Engineering Lab, Biomedical Transport Phenomena, Clinically Inspired Engineering Design, C++ Programming